

Intergalactic Riksbanken Chip Authenticator - README

Group 8

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Intergalactic Riksbanken Chip Authenticator

> A computer vision system for authenticating and valuing intergalactic credit chips (Gold, Silver, Bronze) with simulator, camera, and interactive testing modes.

!Python](<https://www.python.org/>)

!OpenCV](<https://opencv.org/>)

■ Quick Start

Launch All Modes (Recommended)

```
# Install dependencies
pip install -r requirements.txt

# Run the launcher
python launcher.py

# Select mode:
# 1 - Simulator Mode (conveyor belt)
# 2 - Camera Mode (real detection)
# 3 - Interactive Game (manual testing)
```

Individual Modes

Option 1: Simulator Mode (No Camera Required)

```
python main.py
```

Option 2: Camera Mode (Real-time Detection)

```
python camera_main.py
```

Option 3: Interactive Game (Manual Chip Testing)

```
python game.py
```

Features

Simulator Mode

- **Green Conveyor Belt:** 50% screen width, centered with moving texture
- **Three Chip Types:**
 - **GOLD** (Yellow): Value = 3 digits × 10 (e.g., 752 → 7520 CR)
 - **SILVER** (Blue): Value = 3 digits (e.g., 756 → 756 CR)
 - **BRONZE** (Orange): Value = 2 digits × × (e.g., 2×4 → 8 CR)
- **Real & Fake Detection:** 80% real chips, 20% fake chips
- **Straight Line Movement:** Chips move perpendicular to belt motion
- **Real-time Statistics:** Track total value, real/fake counts

Camera Mode

Simulator Mode (`main.py`)

Key	Action
S	Spawn single chip
B	Burst spawn (5 chips)
C	Clear all chips
P	Pause/Resume
R	Reset statistics
Q	Quit

Camera Mode (`camera\_main.py`)

Key	Action

Controls

Key	Action
S	Spawn single chip
B	Burst spawn (5 chips)
C	Clear all chips
P	Pause/Resume

■■■ camera_main.py # Camera detection system

R	Reset statistics
Q	Quit

Project Structure

```
chip_system/
■■■ launcher.py          # Main launcher (all modes)
■■■ main.py              # Conveyor simulator
■■■ camera_main.py       # Camera detection
■■■ game.py              # Interactive game
■■■ requirements.txt      # Dependencies
■■■ .gitignore           # Git ignore rules
■■■ README.md            # This file
■■■ QUICKSTART.md        # Quick reference
■■■ CHANGELOG.md         # Version history
■■■ LICENSE              # MIT License
■■■ CONTRIBUTING.md      # Guidelines
■■■ assets/              # Chip images
■   ■■■ gold.png
■   ■■■ silver.png
■   ■■■ bronze.png
■■■ docs/                # Documentation
■■■ CAMERA_USAGE.md
■■■ MIGRATION.md
```

System Requirements

- Python 3.8+
- OpenCV 4.8+
- NumPy 1.24+
- Webcam (optional, for camera mode)

Value Calculation Rules

Based on STB600 Final Project 2025:

Chip Type	Pattern	Calculation	Example
GOLD	3 digits	$(d_1 \times 100 + d_2 \times 10 + d_3) \times 10$	752 → 7520 CR
SILVER	3 digits	$d_1 \times 100 + d_2 \times 10 + d_3$	756 → 756 CR
BRONZE	2 digits	$d_1 \times d_2$	2×4 → 8 CR

Fake chips have 0 value and are marked in red.

Usage Example

```
from main import ConveyorSimulator

# Create simulator
sim = ConveyorSimulator(width=1280, height=720, conveyor_speed=3)

# Run
sim.run()
```

Credits

Created by Group 8:

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- **Abhishek**

STB600 Final Project 2025

Computer Vision & Image Processing